

MUHAMMAD REHMAN

Full Stack Developer (MERN)

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PROFESSIONAL SUMMARY

Full-stack developer with 1 year of professional experience building production-grade web applications using the MERN stack. Strong expertise in **React.js**, **Next.js**, **Node.js**, **MongoDB**, and **CI/CD automation** using GitHub Actions. Proven ability to reduce production errors by 35%, cut deployment time by 50%, and deliver secure, scalable applications from schema design to deployment.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), TypeScript, Node.js, React.js, Next.js, Express.js, HTML5, CSS3

Databases & ORMs: MongoDB, PostgreSQL, MySQL, Mongoose, Prisma

UI & Styling: Tailwind CSS, Bootstrap, ShadCN, Styled Components

DevOps & Tools: Git, GitHub Actions (CI/CD), Vercel, Render, AWS EC2, Postman, Swagger

AI & Computer Vision: Python, YOLOv8 (Ultralytics), OpenCV, FastAPI, Uvicorn, OpenAI API

PROFESSIONAL EXPERIENCE (1 year)

Associate MERN Stack Developer

Jan 2026 – Present

PixelPK

Lahore, Pakistan

- Refactored and modularized legacy frontend and backend codebases, reducing production errors by **35%** and improving maintainability.
- Implemented end-to-end CI/CD pipelines using GitHub Actions, cutting manual deployment effort by **50%**.
- Built data-driven dashboards and personalized backend workflows in collaboration with design and analytics teams.

Technologies Used: React.js, Node.js, Express.js, MongoDB, GitHub Actions, Tailwind CSS

MERN Stack Intern

Sep 2025 – Dec 2025

PixelPK

Lahore, Pakistan

- Translated Figma designs into pixel-perfect, fully responsive React and Next.js interfaces with lazy loading and reusable component libraries.
- Built full-stack features from MongoDB schema design to REST APIs and implemented JWT-based authentication with role-based access control.

Technologies Used: React.js, Next.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS

PROJECTS

Zewar House , AI-Powered Full-Stack E-Commerce Platform | [GitHub](#)

Oct 2025 – May 2026

Production-grade Pakistani jewellery e-commerce platform with **3 React/Node apps**, **29 REST endpoints**, and an AI Virtual Try-On microservice covering the complete retail lifecycle from catalogue to payment to delivery.

- Built an AI Virtual Try-On system: Python/FastAPI microservice uses **MediaPipe FaceMesh (468 landmarks)** and a trained **RandomForest classifier** to detect 6 face shapes and generate real-time jewellery overlays via OpenCV alpha blending personalised recommendations delivered per face shape.
- Architected a **3-app MERN platform** (customer storefront, admin dashboard, AI server) with JWT multi-role auth, **OTP email verification** (Nodemailer + MongoDB 10-min TTL), and **Cloudinary CDN** for multi-image product management with automatic file cleanup.
- Implemented end-to-end e-commerce flow: **Stripe payments (PKR)**, cash on delivery, 5-stage order tracking, wishlist, star-rating review system, and search/filter across 4 jewellery categories × 5 materials.

Technologies Used: React.js, Node.js, Express.js, MongoDB, Python, FastAPI, MediaPipe, scikit-learn, Stripe, Cloudinary, TailwindCSS, JWT, Nodemailer

AI-Powered Smart Meter Reading System | [GitHub](#)

Sep 2025 – May 2026

End-to-end utility management platform that auto-extracts electricity meter readings from photographs using two fine-tuned **YOLOv8** object detection models, served by a full **TypeScript + React + PostgreSQL** stack with dedicated dashboards for Admin, Field Staff, and Consumer roles.

- Trained two YOLOv8 models on **11,000+ real-world meter images** one for meter ROI cropping, one for digit recognition and engineered a **7-variant OpenCV preprocessing pipeline** (CLAHE, bilateral filtering, gamma correction, unsharp masking, morphological ops) with automatic confidence-score selection, achieving reliable reading extraction across varied lighting and field conditions.
- Built a **confidence-based hybrid AI + human review workflow**: low-confidence predictions are automatically flagged for administrator approval before billing, balancing automation with accuracy in a zero-error-tolerance domain.
- Delivered a production-grade **3-tier architecture** (React + Vite frontend, Express REST API with 12+ route groups, FastAPI AI inference server) with JWT auth, role-based access control, automated monthly billing engine, cron-scheduled background jobs, dispute management, audit logging, and PDF/CSV export deployed on **Vercel** with **Cloudinary** CDN for image storage.

Full-featured job portal with role-based JWT authentication, job posting, filtering, and application tracking for employers and candidates.

- Designed MongoDB schemas, centralised error handling with Winston logger; deployed backend on AWS EC2.

EDUCATION

Bachelor of Science in Software Engineering | Jun 2026 | University of Lahore CGPA: 3.93 / 4.0